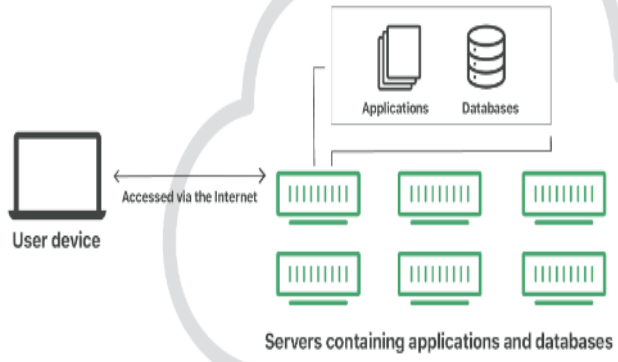
Unit – I
Introduction to Cloud Computing

1.1.1 cloud computing
About cloud
About computing
Definition of cloud computing
Central Ideas Behind Cloud Computing

1.1.2 Characteristics of Cloud Computing
1.1.3 Challenges of Cloud Computing
1.1.4 Novel Applications of cloud computing
1.1.5 Security risk of cloud computing
1.2.1 Advantages and Disadvantages of Cloud Computing

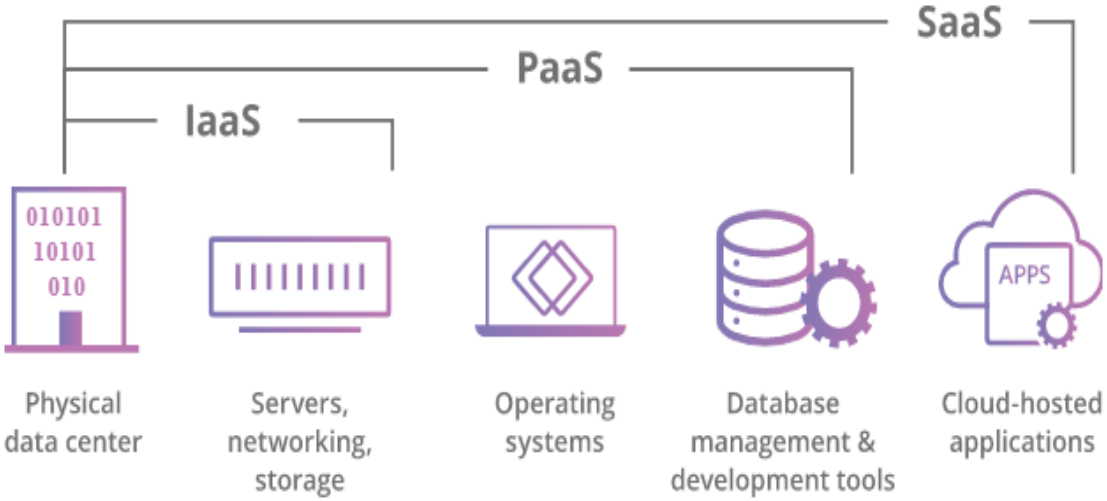
1.1.1 Cloud Computing :

- **About cloud :**
- ✓ "The cloud" refers to servers that are accessed over the Internet, and the software and databases that run on those servers.
- ✓ Cloud servers are located in **data centers** all over the world. By using cloud computing, users and companies do not have to manage physical servers themselves or run software applications on their own machines.
- ✓ **A data center** is a physical location that stores computing machines and their related hardware equipment.



The Cloud

- ✓ **Cloud computing** can also be thought of as utility computing or on-demand computing. The name cloud computing was inspired by the cloud symbol that's often used to represent the internet in flowcharts and diagrams.
- ✓ The resources available in the cloud are known as "services," since they are actively managed by a cloud provider.
- ✓ Cloud services include infrastructure, applications, development tools, and data storage, among other products.
- ✓ These services are sorted into several different categories, or *service models* as shown in the figure below.



❖ **About computing:**

- ✓ In cloud computing, the term "compute" describes concepts and objects related to software computation.
- ✓ It is a generic term used to reference processing power, memory, networking, storage, and other resources required for the computational success of any program.

- ✓ **For example**, applications that run machine learning algorithms or 3D graphics rendering functions require many gigs of RAM and multiple CPUs to run successfully. In this case, the CPUs, RAM, and Graphic Processing Units required will be called compute resources, and the applications would be compute-intensive applications.

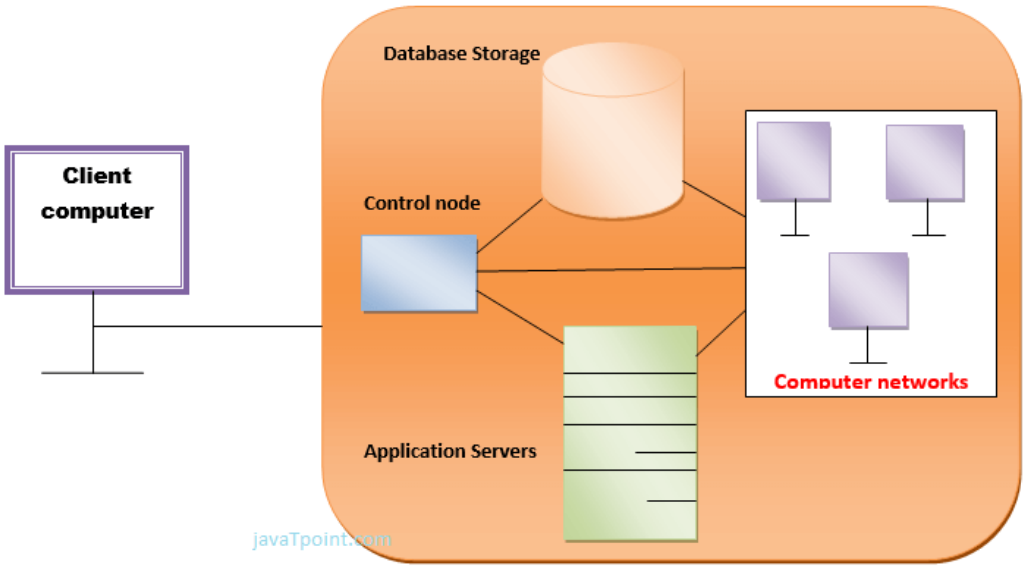
❖ **Definition of cloud computing :**

- ✓ **Cloud computing** is the on-demand availability of computer system resources, especially data storage (cloud storage) and computing power, without direct active management by the user.
- ✓ Large clouds often have functions distributed over multiple locations, each of which is a data center. Cloud computing relies on sharing of resources to achieve coherence and typically

uses a pay-as-you-go model, which can help in reducing capital expenses but may also lead to unexpected operating expenses for users.

- ✓ Cloud computing refers to the use of hosted services, such as data storage, servers, databases, networking, and software over the internet.
- ✓ The data is stored on physical servers, which are maintained by a cloud service provider. Computer system resources, especially data storage and computing power, are available on-demand, without direct management by the user in cloud computing.

❖ Central Ideas Behind Cloud Computing:

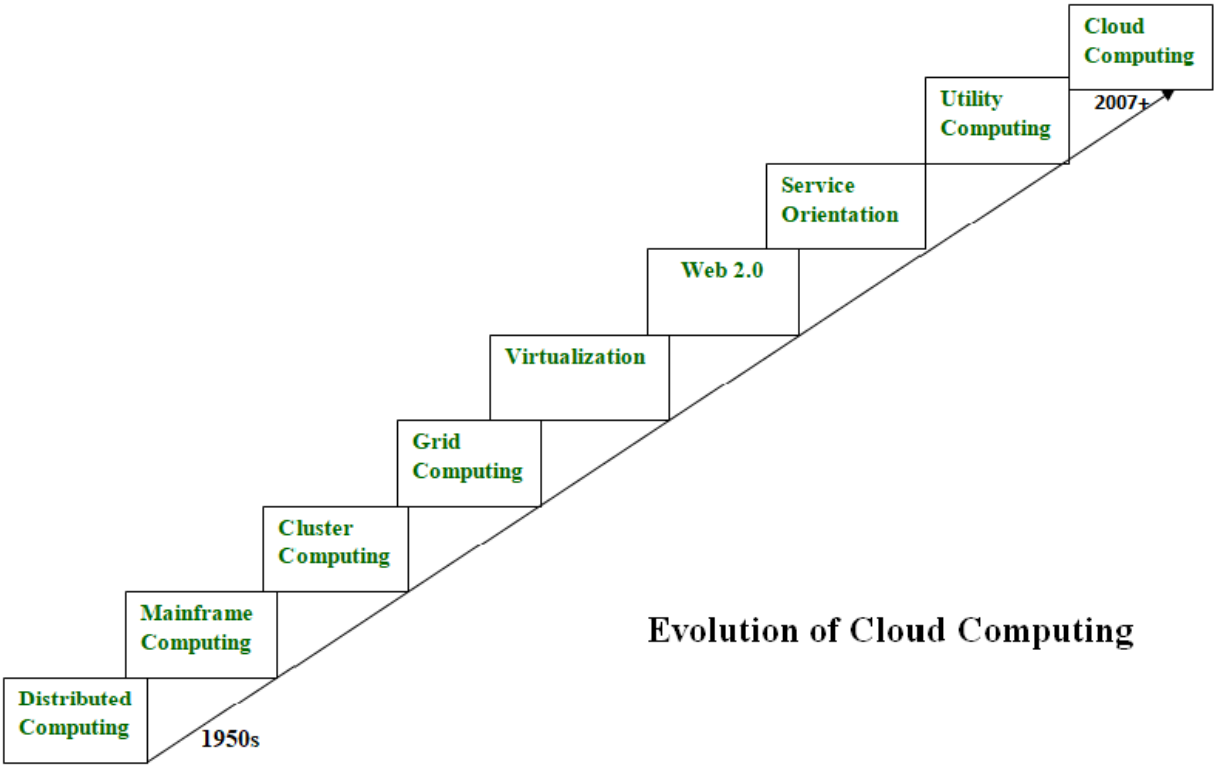


- ✓ In a cloud computing system, there is a significant workload shift. Local computers have no longer to do all the heavy lifting when it comes to run applications. But cloud computing can handle that much heavy load easily and automatically. Hardware and software demands on the user's side decrease. The only thing the user's computer requires to be able to run is the cloud computing interface software of the system, which can be as simple as a Web browser and the cloud's network takes care of the rest.

❖ History:

Cloud computing is all about renting computing services.

This idea first came in the 1950s. In making cloud computing what it is today, following technologies played a vital role.



- **Distributed Systems:**

✓ It is a composition of multiple independent systems but all of them are depicted as a single entity to the users.
✓ The purpose of distributed systems is to share resources and also use them effectively and efficiently.
✓ Distributed systems possess characteristics such as scalability, concurrency, continuous availability, heterogeneity, and independence in failures.
✓ But the main problem with this system was that all the systems were required to be present at the same geographical location.
✓ Thus to solve this problem, distributed computing led to three more types of computing and they were-Mainframe computing, cluster computing, and grid computing.

- **Mainframe computing:**

✓ Mainframes which first came into existence in 1951 are highly powerful and reliable computing machines.
✓ These are responsible for handling large data such as massive input-output operations.
✓ Even today these are used for bulk processing tasks such as online transactions etc.
✓ These systems have almost no downtime with high fault tolerance. After distributed computing, these increased the processing capabilities of the system.
✓ But these were very expensive. To reduce this cost, cluster computing came as an alternative to mainframe technology.

- **Cluster computing:**

- ✓ In 1980s, cluster computing came as an alternative to mainframe computing.
- ✓ Each machine in the cluster was connected to each other by a network with high bandwidth.
- ✓ These were way cheaper than those mainframe systems. These were equally capable of high computations.
- ✓ Also, new nodes could easily be added to the cluster if it was required. Thus, the problem of the cost was solved to some extent but the problem related to geographical restrictions still pertained.
- ✓ To solve this, the concept of grid computing was introduced.

- **Grid computing:**

- ✓ In 1990s, the concept of grid computing was introduced.
- ✓ It means that different systems were placed at entirely different geographical locations and these all were connected via the internet.
- ✓ These systems belonged to different organizations and thus the grid consisted of heterogeneous nodes.
- ✓ Although it solved some problems but new problems emerged as the distance between the nodes increased.
- ✓ The main problem which was encountered was the low availability of high bandwidth connectivity and with it other network associated issues. Thus. cloud computing is often referred to as "Successor of grid computing".

- **Virtualization:**

- ✓ It was introduced nearly 40 years back. It refers to the process of creating a virtual layer over the hardware which allows the user to run multiple instances simultaneously on the hardware.
- ✓ It is a key technology used in cloud computing. It is the base on which major cloud computing services such as Amazon EC2, VMware vCloud, etc work on.
- ✓ Hardware virtualization is still one of the most common types of virtualization.

- **Web 2.0:**

- ✓ It is the interface through which the cloud computing services interact with the clients.
- ✓ It is because of Web 2.0 that we have interactive and dynamic web pages. It also increases flexibility among web pages.
- ✓ Popular examples of web 2.0 include Google Maps, Facebook, Twitter, etc. Needless to say, social media is possible because of this technology only. It gained major popularity in 2004.

- **Service orientation:**

- ✓ It acts as a reference model for cloud computing. It supports low-cost, flexible, and evolvable applications.

- ✓ Two important concepts were introduced in this computing model. These were Quality of Service (QoS) which also includes the SLA (Service Level Agreement) and Software as a Service (SaaS).

- **Utility computing:**

- ✓ It is a computing model that defines service provisioning techniques for services such as computer services along with other major services such as storage, infrastructure, etc which are provisioned on a pay-per-use basis.

1.1.2 Characteristics of Cloud Computing

❖ **Following are the characteristics of Cloud Computing:**

1. Resources Pooling

This means that the Cloud provider used a multi-leaner model to deliver the computing resources to various customers. There are various allocated and reassigned physical and virtual resources, which rely on customer demand. In general, the customer has no control or information about the location of the resources provided, but can choose location on a higher level of abstraction.

2. On-Demand Self-Service

This is one of the main and useful advantages of Cloud Computing as the user can track server uptimes, capability and network storage on an ongoing basis. The user can also monitor computing functionalities with this feature.

3. Easy Maintenance

The servers are managed easily and the downtime is small and there are no downtime except in some cases. Cloud Computing offers an update every time that increasingly enhances it. The updates are more system friendly and operate with patched bugs faster than the older ones.

4. Large Network Access

The user may use a device and an Internet connection to access the cloud data or upload it to the cloud from anywhere. Such capabilities can be accessed across the network and through the internet.

5. Availability

The cloud capabilities can be changed and expanded according to the usage. This review helps the consumer to buy additional cloud storage for a very small price, if necessary.

6. Automatic System

Cloud computing analyzes the data required automatically and supports a certain service level of measuring capabilities. It is possible to track, manage and report the usage. It provides both the host and the customer with accountability**.**

7. Economical

VPMP POLYTECHNIC

It is a one-off investment since the company (host) is required to buy the storage, which can be made available to many companies, which save the host from monthly or annual costs. Only the amount spent on the basic maintenance and some additional costs are much smaller.

8. Security

Cloud Security is one of cloud computing's best features. It provides a snapshot of the data stored so that even if one of the servers is damaged, the data cannot get lost. The information is stored on the storage devices, which no other person can hack or use. The service of storage is fast and reliable**.**

9. Pay as you go

Users only have to pay for the service or the space in cloud computing. No hidden or additional charge to be paid is liable to pay. The service is economical and space is often allocated free of charge.

10. Measured Service

Cloud Computing resources that the company uses to monitor and record. This use of resources is analyzed by charge-per-use capabilities. This means that resource use can be measured and reported by the service provider, either on the virtual server instances running through the cloud. You will receive a models pay depending on the manufacturing company's actual consumption.

1.1.3 Challenges of Cloud Computing

The most common challenges that are faced when dealing with cloud computing are:

1. Data Security and Privacy

Data security is a major concern when switching to cloud computing. User or organizational data stored in the cloud is critical and private. Even if the cloud service provider assures data integrity, it is your responsibility to carry out user authentication and authorization, identity management, data encryption, and access control. Security issues on the cloud include identity theft, data breaches, malware infections, and a lot more which eventually decrease the trust amongst the users of your applications. This can in turn lead to potential loss in revenue alongside reputation and stature. Also, dealing with cloud computing requires sending and receiving huge amounts of data at high speed, and therefore is susceptible to data leaks.

2. Cost Management

Even as almost all cloud service providers have a “Pay As You Go” model, which reduces the overall cost of the resources being used, there are times when there are huge costs incurred to the enterprise using cloud computing. When there is under optimization of the resources, let’s

say that the servers are not being used to their full potential, add up to the hidden costs. If there is a degraded application performance or sudden spikes or overages in the usage, it adds up to the overall cost. Unused resources are one of the other main reasons why the costs go up. If you turn on the services or an instance of cloud and forget to turn it off during the weekend or when there is no current use of it, it will increase the cost without even using the resources.

3. Multi-Cloud Environments

Due to an increase in the options available to the companies, enterprises not only use a single cloud but depend on multiple cloud service providers. Most of these companies use hybrid cloud tactics and close to 84% are dependent on multiple clouds. This often ends up being hindered and difficult to manage for the infrastructure team. The process most of the time ends up being highly complex for the IT team due to the differences between multiple cloud providers.

4. Performance Challenges

Performance is an important factor while considering cloud-based solutions. If the performance of the cloud is not satisfactory, it can drive away users and decrease profits. Even a little latency while loading an app or a web page can result in a huge drop in the percentage of users. This latency can be a product of inefficient load balancing, which means that the server cannot efficiently split the incoming traffic so as to provide the best user experience. Challenges also arise in the case of fault tolerance, which means the operations continue as required even when one or more of the components fail.

5. Interoperability and Flexibility

When an organization uses a specific cloud service provider and wants to switch to another cloud-based solution, it often turns up to be a tedious procedure since applications written for one cloud with the application stack are required to be re-written for the other cloud. There is a lack of flexibility from switching from one cloud to another due to the complexities involved. Handling data movement, setting up the security from scratch and network also add up to the issues encountered when changing cloud solutions, thereby reducing flexibility.

6. High Dependence on Network

Since cloud computing deals with provisioning resources in real-time, it deals with enormous amounts of data transfer to and from the servers. This is only made possible due to the availability of the high-speed network. Although these data and resources are exchanged over the network, this can prove to be highly vulnerable in case of limited bandwidth or cases

when there is a sudden outage. Even when the enterprises can cut their hardware costs, they need to ensure that the internet bandwidth is high as well there are zero network outages, or else it can result in a potential business loss. It is therefore a major challenge for smaller enterprises that have to maintain network bandwidth that comes with a high cost.

7. Lack of Knowledge and Expertise

Due to the complex nature and the high demand for research working with the cloud often ends up being a highly tedious task. It requires immense knowledge and wide expertise on the subject. Although there are a lot of professionals in the field they need to constantly update themselves. Cloud computing is a highly paid job due to the extensive gap between demand and supply. There are a lot of vacancies but very few talented cloud engineers, developers, and professionals. Therefore, there is a need for upskilling so these professionals can actively understand, manage and develop cloud-based applications with minimum issues and maximum reliability.

1.1.4 Novel Applications of cloud computing

- ✓ How is the cloud actually used? modern public clouds have been applied across countless use cases, such as the following:
- ✓ **Testing and development.** Ready-made, tailored environments can expedite timelines and milestones.
- ✓ **Production workload hosting.** Organizations are using the public cloud to host live production workloads. This requires careful design and architecture of cloud resources and services needed to create an adequate operational environment for the workload and its required level of resilience.
- ✓ Big data analytics. Remote data centers through cloud storage are flexible and scalable and can provide valuable data-driven insights. Major cloud providers offer services tailored to big data analytics and projects, such as Amazon EMR and Google Cloud Dataproc.
- ✓ **IaaS.** IaaS lets companies host IT infrastructures and access compute, storage and network capabilities in a scalable manner. Pay-as-you-go subscription models are cost-effective, as they can help companies save on upfront IT costs.

- ✓ **PaaS.** PaaS can help companies develop, run and manage applications more easily and flexibly, at a lower cost than maintaining a platform on premises. PaaS services can also increase the development speed for applications and enable higher-level programming.
- ✓ **Hybrid cloud.** Organizations have the option to use the appropriate cloud -- private or public -- for different workloads and applications to optimize cost and efficiency according to the circumstance.
- ✓ **Multi-cloud.** Using multiple different cloud services from separate cloud providers can help subscribers find the best cloud service fit for diverse workloads with specific requirements.
- ✓ **Storage.** Large amounts of data can be stored remotely and accessed easily. Clients only have to pay for storage that they actually use.
- ✓ **Disaster recovery.** Cloud offers faster recovery than traditional on-premises DR. Furthermore, it's offered at lower costs.
- ✓ **Data backup.** Cloud backup options are generally easier to use. Users don't have to worry about availability and computing capacity, and the cloud provider manages data security.
- ✓ **Artificial intelligence as a service.** Cloud computing lets individuals without formal knowledge or expertise in data sciences reap the benefits of AIaaS. For example, a web developer might create a facial recognition app with their web development skills. AI is available as a service in the cloud and accessible via the API. This lets users automate routine tasks, saving time and personnel costs. Businesses can also enhance decision-making by using AI to predict outcomes based on historical datasets.
- ✓ **Internet of things.** Cloud computing simplifies the processing and management of data from IoT devices. Cloud platforms offer the scalability and processing capacity required to handle the enormous amounts of data produced by IoT devices, facilitating real-time analytics and decision-making. For example, an IoT device system such as Google Nest or Amazon Alexa can collect data on how much energy is used inside a smart home. The

device can then use cloud computing to analyze the gathered data and make recommendations to the homeowner on how to reduce energy consumption.

1.1.5 Security risk of cloud computing

Some most common Security Risks of Cloud Computing are given below-

1. **Data Loss:**

   Data loss is the most common cloud security risks of cloud computing. It is also known as data leakage. Data loss is the process in which data is being deleted, corrupted, and unreadable by a user, software, or application. In a cloud computing environment, data loss occurs when our sensitive data is somebody else's hands, one or more data elements can not be utilized by the data owner, hard disk is not working properly, and software is not updated.

2. **Hacked Interfaces and Insecure APIs:**

   As we all know, cloud computing is completely depends on Internet, so it is compulsory to protect interfaces and APIs that are used by external users. APIs are the easiest way to communicate with most of the cloud services. In cloud computing, few services are available in the public domain. These services can be accessed by third parties, so there may be a chance that these services easily harmed and hacked by hackers.

3. **Data Breach:**

   Data Breach is the process in which the confidential data is viewed, accessed, or stolen by the third party without any authorization, so organization's data is hacked by the hackers.

4. **Vendor lock-in:**

   Vendor lock-in is the of the biggest security risks in cloud computing. Organizations may face problems when transferring their services from one vendor to another. As different vendors provide different platforms, that can cause difficulty moving one cloud to another.

5. **Increased complexity strains IT staff:**

Migrating, integrating, and operating the cloud services is complex for the IT staff. IT staff must require the extra capability and skills to manage, integrate, and maintain the data to the cloud.

6. **Spectre & Meltdown:**

Spectre & Meltdown allows programs to view and steal data which is currently processed on computer. It can run on personal computers, mobile devices, and in the cloud. It can store the password, your personal information such as images, emails, and business documents in the memory of other running programs.

7. **Denial of Service (DoS) attacks:**

Denial of service (DoS) attacks occur when the system receives too much traffic to buffer the server. Mostly, DoS attackers target web servers of large organizations such as banking sectors, media companies, and government organizations. To recover the lost data, DoS attackers charge a great deal of time and money to handle the data.

8. **Account hijacking:**

Account hijacking is a serious security risk in cloud computing. It is the process in which individual user's or organization's cloud account (bank account, e-mail account, and social media account) is stolen by hackers. The hackers use the stolen account to perform unauthorized activities.

1.2.1 Advantages and Disadvantages of Cloud Computing

❖ **Advantages of Cloud Computing**

✓ **Cost efficiency**

✓ The biggest reason behind companies shifting to Cloud Computing is that it takes considerably lesser cost than any on-premise technology. Now, companies need not store data in disks anymore as the cloud offers enormous storage space, saving money and resources.

✓ **High speed**

✓ Cloud Computing lets us deploy the service quickly with fewer clicks. This quick deployment lets us get the resources required for our system within minutes.

- ✓ **Excellent accessibility**

- ✓ Storing information in the cloud allows us to access it anywhere and anytime, regardless of the machine making it a highly accessible and flexible technology of the present time.

- ✓ **Back-up and restore data**

- ✓ Once data is stored in the cloud, it is easier to get its backup and recovery, which is quite a time-consuming process in on-premise technology.

- ✓ **Manageability**

- ✓ Cloud Computing eliminates the need for IT infrastructure updates and maintenance since the service provider ensures timely, guaranteed, and seamless delivery of our services and also takes care of all the maintenance and management of our IT services according to the service-level agreement (SLA).

- ✓ **Sporadic batch processing**

- ✓ Cloud computing lets us add or subtract resources and services according to our needs. So, if the workload is not 24/7, we need not worry about the resources and services getting wasted, and we won't end up stuck with unused services.

- ✓ **Strategic Edge**

- ✓ Cloud computing provides a company with a competitive edge over its competitors when it comes to accessing the latest and most mission-critical applications that it needs without having to invest its time and money in its installations. It lets the company focus on keeping up with the business competition by offering access to the most trending and in-demand applications and doing all the manual work of installing and maintaining the applications for the company.

- ✓ **Easy implementation**

- ✓ With cloud hosting, businesses can retain the same applications and processes without having to deal with the backend technicalities. Cloud infrastructure can be accessed easily.

- ✓ **No hardware required**
- ✓ Since the cloud hosts everything, there is no requirement for a physical storage center. However, a backup is recommended for disaster management and business continuity.

- ✓ **Automatic Software Integration**
- ✓ In the cloud, software integration occurs automatically. No additional efforts are necessary for the customization and integration of applications to meet preferences.

- ✓ **Reliability**
- ✓ Cloud hosting comes with the biggest advantage of reliability. One doesn't have to worry about changes due to instant updates.

- ✓ **Mobility**
- ✓ Employees working on-premises or from remote locations can easily access all the cloud services as long as they have an internet connection.

- ✓ **Unlimited storage capacity**
- ✓ The cloud comes with an almost limitless storage capacity, which can be expanded at any time for a very nominal monthly fee.

- ✓ **Collaboration**
- ✓ Cloud computing platforms help employees located in different locations to conveniently and securely collaborate with each other.

❖ **Disadvantages of Cloud Computing**

- ✓ **Vulnerability to attacks**
- ✓ Storing data in the cloud may pose serious challenges in terms of information theft since every company's data is online. A security breach is something that even the best organizations have suffered from, and it's a potential risk in the cloud as well. Although advanced security measures are deployed on the cloud, still storing confidential data in the cloud can be a risky affair, and hence vulnerability to attacks shall be considered.

- **Network connectivity dependency**

- Cloud computing is entirely dependent on the Internet. This direct tie-up with the Internet means that a company needs to have reliable and consistent Internet service as well as a fast connection and bandwidth to reap the benefits of cloud computing.

- **Downtime**

- Downtime is considered as one of the biggest potential downsides of using cloud computing. Cloud providers may sometimes face technical outages that can happen due to various reasons, such as loss of power, low Internet connectivity, data centers going out of service for maintenance, etc. This can lead to temporary downtime in the cloud service.

- **Vendor lock-in**

- When a company needs to migrate from one cloud platform to another, it might face some serious challenges because of the differences between vendor platforms. Hosting and running the applications of the current cloud platform on some other platform may cause support issues, configuration complexities, and additional expenses. The company's data might also be left vulnerable to security attacks due to compromises that might have been made during migrations.

- **Limited control**

- Cloud customers may face limited control over their deployments. Cloud services run on remote servers that are completely owned and managed by service providers, which makes it hard for the companies to have the level of control that they would want over their back-end infrastructure.

- **May not get all the features**

- Some cloud providers offer only limited versions with the most popular features. Before signing up, it is important to know what cloud services are provided.

- **Can't do away with servers altogether**

- While there may be fewer servers to handle, it is not recommended to entirely let go of all your servers and staff because redundancy is key for cloud backup and recovery.

- ✓ **No Redundancy**

- ✓ A cloud server is not redundant or backed up. Although it is an extra expense, make sure to invest in a redundancy plan as in most cases, it will be well worth it. Technology does sometimes seem to fail.

- ✓ **Bandwidth issues**

- ✓ For better performance, it is advisable not to pack large numbers of storage devices and servers into a small set of data centers. The additional charges could, however, be significantly costly.

- ✓ **Lacks Support**

- ✓ Cloud computing companies do not provide proper support to their customers and expect them to depend on FAQs or online help.

- ✓ **Varied Performances**

- ✓ In a cloud environment, applications run on the server, which also provides resources to other businesses. This means that any attack or harmful activity by a tenant could affect the performance of the shared resources.

VPMP POL